Software Project Management Plan

EswatiniHomes Property Listing System

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Standard Alignment: IEEE Std 16326 (Software Project Management)

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1 Introduction

1.1 Project Overview

EswatiniHomes is a web-based real estate platform designed for the Eswatini market. It serves as a centralized digital hub where Estate Agents can list properties (Houses, Apartments, Land, Commercial, Townhouses) for Sale or Rent, and where the public can search, filter, and inquire about these properties. The system features a robust Admin dashboard for content moderation and an Agent dashboard for commission tracking.

1.2 Purpose of the Plan

This document defines the strategy, processes, resources, and schedules required to develop and deliver the EswatiniHomes system. It serves as the primary reference for the project team to ensure the project is completed on time, within budget, and to the highest quality standards.

1.3 Project Objectives (SMART Goals)

- **Specific:** Build a fully functional PHP/MySQL property listing web application.
- **Measurable:** Achieve 100% pass rate on all 50 predefined test cases.
- **Achievable:** Leverage open-source tools (XAMPP, phpMyAdmin, VS Code) to minimize costs.
- **Relevant:** Solve the problem of fragmented property listings in Eswatini.
- **Time-bound:** Deliver the final product and documentation within 6 weeks.

2 Project Scope & Methodology

2.1 Product Scope (In-Scope)

- **Public Portal:** Homepage, dynamic search engine (filter by type, listing type, max price), property details page, and contact/lead form.
- **Authentication System:** Secure login/registration using password hashing (bcrypt).
- **Agent Portal:** Ability to upload property images, manage their own listings, and log closed deals (calculating commissions).
- **Admin Portal:** Approve/Reject pending listings, delete properties (including images from the server), and manage user roles.
- **Database:** Relational MySQL database with 6 tables (`properties`, `agents`, `users`, `admins`, `leads`, `agent_deals`).

2.2 Out-of-Scope

- Native iOS/Android mobile applications.
- Live chat/messaging system between users and agents.
- Automated payment gateways for premium listings.
- Automated email/SMS notifications (v1.0 will rely on database logging).

2.3 Software Development Life Cycle (SDLC) Model

For the development of the EswatiniHomes system, the **Iterative and Incremental Development Model** (with Agile principles) will be utilized.

Why this model was chosen:

- **Evolving Requirements:** Real estate platforms often require visual tweaks and database adjustments. A strict Waterfall model would make this difficult.
- **Early Feedback:** By building the system in iterations, the client can review progress early and request changes before final delivery.
- **Risk Mitigation:** Technical risks (like XAMPP port conflicts or file permission errors) are identified and resolved in early iterations.

The Iterative Phases:

1. **Iteration 1 (Foundation):** Planning, SDD, and Database Schema creation.
2. **Iteration 2 (Core):** Authentication, `config.php`, and the public `index.php` search page.
3. **Iteration 3 (Management):** Admin dashboard, Agent dashboards, and secure `delete.php`.
4. **Iteration 4 (Refinement):** Testing, Security hardening (CSRF, SQL Injection), and deployment.

3 Work Breakdown Structure (WBS)

4 Resource Management

4.1 Hardware Resources

- **Developer Workstation:** Windows 10/11, Intel Core i5 (or AMD equivalent), 8GB+ RAM, 256GB SSD.
- **Server (Production):** Shared hosting or VPS with cPanel, PHP 8.x, MySQL 8.x.

Phase	Task Description	Week
1. Planning	Requirements gathering, SRS creation, Environment Setup.	Week 1
2. Design	Database schema design (ERD), SDD creation, UI prototyping.	Week 2
3. Development	Core backend (PHP), frontend (HTML/CSS), Auth system, Dashboards.	Weeks 3-4
4. Testing	Unit testing, integration testing, security testing, bug fixing.	Week 5
5. Deployment	Server setup, database import, User Acceptance Testing (UAT).	Week 6

Table 1: High-Level Work Breakdown Structure

4.2 Software Resources

- **Local Server:** XAMPP (installed in `C:\xampp_2` to avoid system conflicts).
- **Code Editor:** Visual Studio Code (with PHP IntelliSense).
- **Database UI:** phpMyAdmin.
- **Version Control:** Git (GitHub repository).
- **Design Tools:** draw.io (ERDs), Figma (UI mockups).

5 Risk Management Plan

ID	Risk Description	Prob.	Impact	Mitigation Strategy
R01	**Port Conflict:** MySQL port 3306 blocked.	High	High	Change XAMPP MySQL port to 3307 and update `my.ini`.
R02	**File Permission:** UAC "Access Denied".	High	High	Run `xampp-control.exe` as Administrator.
R03	**Data Loss:** Accidental database drop.	Low	Critical	Daily exports of `database.sql`; Git commits for code.
R04	**Security Breach:** SQL Injection/XSS.	Med	High	Use Prepared Statements and `htmlspecialchars()`.
R05	**Scope Creep:** New features requested.	High	Med	Log requests for "Phase 2" release; stick to SRS.

Table 2: Risk Management Matrix

6 Quality Assurance & Testing Strategy

To ensure the final product conforms to the SRS, the following testing protocols will be executed:

1. **Static Code Analysis:** Review PHP code for syntax errors and compatibility with PHP 8.x.
2. **Unit Testing:** Verify `currency()` formats numbers correctly; verify `verify_csrf_token()` rejects invalid tokens.
3. **Integration Testing:** Test `delete.php` to ensure it deletes the database row *and* the physical image file.
4. **Security Testing:** Attempt to access `admin/dashboard.php` without login; enter SQL injection payloads in the search bar.
5. **User Acceptance Testing (UAT):** Client performs a checklist of 20 core tasks (e.g., "Add a property", "Search for a house").

7 Deployment & Maintenance Plan

7.1 Deployment Strategy

1. **Freeze Code:** No new features added in Week 6.
2. **Export Database:** Generate final `database.sql` from local phpMyAdmin.
3. **Upload Files:** Upload the `/property-listing` folder to `public_html`.
4. **Database Import:** Create live MySQL database and import `database.sql`.
5. **Configuration:** Update `include/config.php` with live host credentials.
6. **Smoke Test:** Run final checks on the live URL.

7.2 Maintenance Plan

- **Bug Fixes:** 30-day warranty for critical bugs.
- **Backups:** Daily automated backups via cPanel or cron job.
- **Updates:** Check for PHP/MySQL security patches every 6 months.